SCHEDULING ALGORITHMS 241401097

1)FIRST COME FIRST SERVE(FCFS)

Code

```
#include <stdio.h>

int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);

        int at[n], bt[n], ct[n], tat[n], wt[n];

        for(int i=0; i<n; i++){
                printf("Enter Arrival time and Burst Time for P%d: ", i+1);
                scanf("%d %d:",&at[i],&bt[i]);
        }

        float total_tat = 0, total_wt = 0;

        ct[0] = at[0] + bt[0];

        for(int i=0; i<n; i++){
                if(ct[i-1]<at[i]){
                        ct[i] = at[i] + bt[i];
                } else {
                        ct[i] = ct[i-1] + bt[i];
                }
        }

        for(int i=0; i<n; i++){
                tat[i] = ct[i] - at[i];
                wt[i] = tat[i] - bt[i];

                total_tat += tat[i];
                total_wt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f", total_tat/n);
        printf("\nAverage Waiting Time: %.2f", total_wt/n);

        return 0;
}
```

Output:-

```
[user123@localhost scheduling]$ gcc fcfs.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes: 4
Enter Arrival time and Burst Time for P1: 0 2
Enter Arrival time and Burst Time for P2: 1 2
Enter Arrival time and Burst Time for P3: 5 3
Enter Arrival time and Burst Time for P4: 6 4

Average Turnaround Time: 3.50
Average Waiting Time: 0.75[user123@localhost scheduling]$ cat fcfs.c
```

2)SHORTEST JOB FIRST(SJF)

```c
#include <stdio.h>

int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);

        int at[n], bt[n], ct[n], tat[n], wt[n];
        int done[n];

        for(int i=0; i<n; i++){
                printf("Enter Arrival Time and Burst Time for P%d: ",i+1);
                scanf("%d %d",&at[i], &bt[i]);
                done[i] = 0;
        }

        int completed = 0;
        int completionTime = 0;
        float avgtat = 0, avgwt = 0;

        while(completed < n){
                int idx = -1;
                int minBT = 9999;

                for(int i=0; i<n; i++){
                        if(at[i]<=completionTime && done[i] == 0){
                                if(bt[i]<minBT){
                                        minBT = bt[i];
                                        idx = i;
                                }
                        }
                }

                if(idx != -1){
                        if(completionTime<at[idx]){
                                completionTime = at[idx];
                        }
                        completionTime += bt[idx];
```